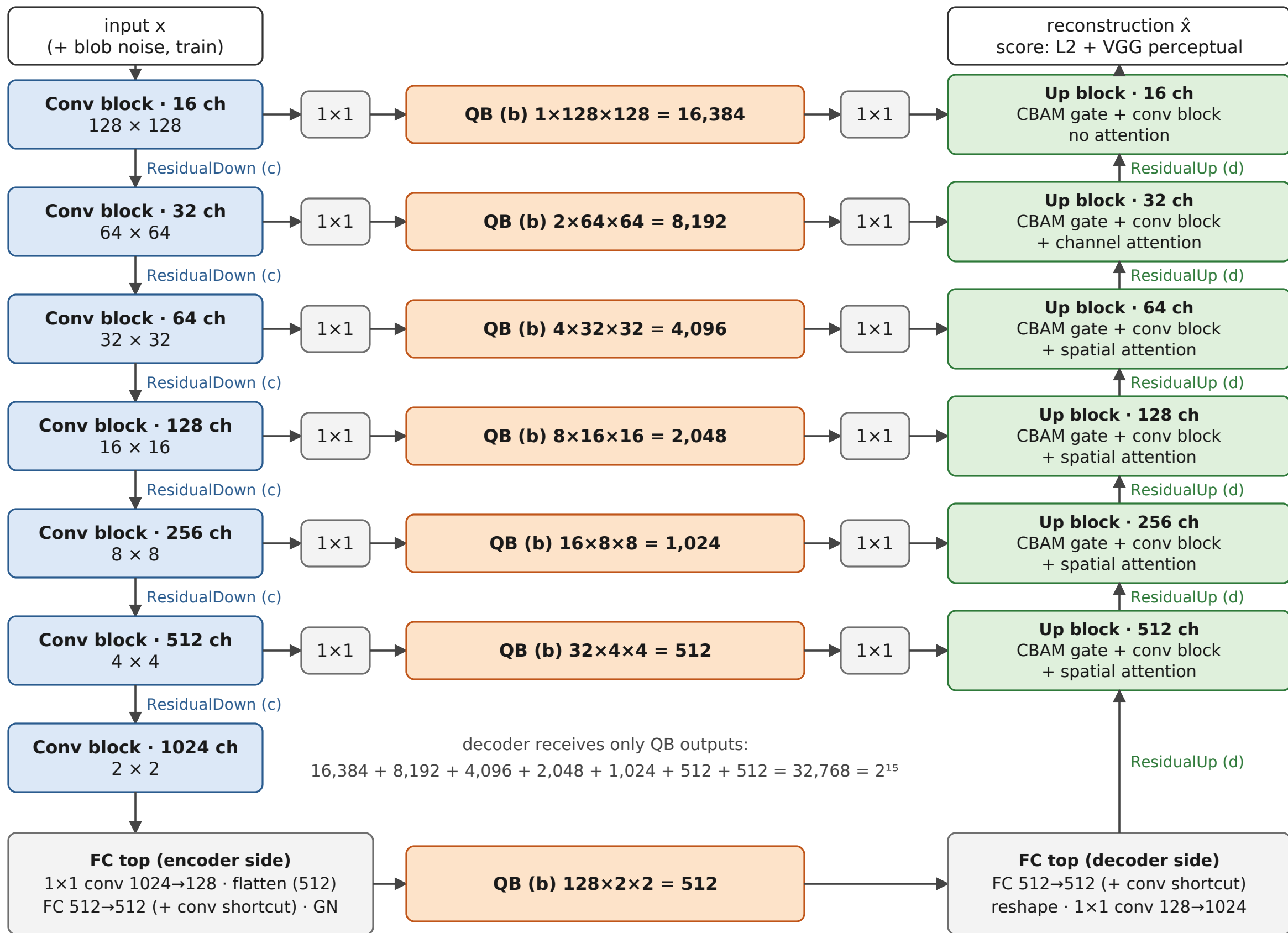
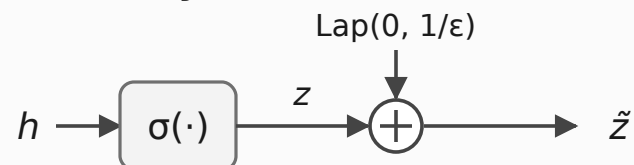


(a) Overall architecture (7 levels, 128 × 128 input)

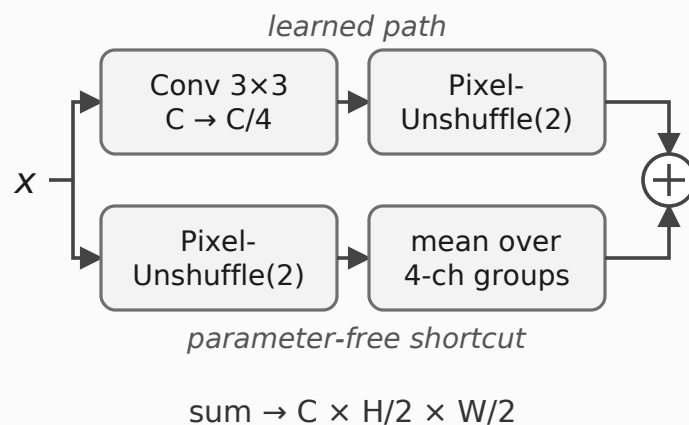


(b) QB layer



train: $\tilde{z} = \sigma(h) + n$, $n \sim \text{Lap}(0, 1/\epsilon)$
test: $\tilde{z} = \sigma(h)$ or $1[\sigma(h) > 0.5]$
 ϵ -local DP per channel ($\Delta = 1$);
Heaviside at test: ≤ 1 bit / channel

(c) ResidualDown $C \times H \times W \rightarrow C \times H/2 \times W/2$



(d) ResidualUp $C \times H \times W \rightarrow C' \times 2H \times 2W$

